

Digital Circuits Quiz 2

1. In a 3-to-8 decoder, how many outputs are active for a given 3-bit input?
 - (A) 1
 - (B) 2
 - (C) 4
 - (D) 8
2. A 2-to-4 decoder with an enable input is often used in memory selection. If the enable signal is low, what will be the state of the outputs?
 - (A) All outputs are active
 - (B) All outputs are inactive
 - (C) The outputs will depend on the inputs
 - (D) Only one output will be active
3. Construct a 4-to-16 decoder from 3-to-8 decoder(s) and one (exactly) 2-to-4 decoder.
4. How many select lines are required for an 8-to-1 multiplexer?
 - (A) 2
 - (B) 3
 - (C) 4
 - (D) 8
5. If an 8-to-1 multiplexer has inputs I_0 to I_7 , which input will be selected if the select lines are set to 101?
 - (A) I_0 $\&4$
 - (B) I_2
 - (C) I_5
 - (D) I_7
6. Given the Boolean expression:

$$F(A, B, C) = A'B + AC$$

Implement this expression using a 4-to-1 multiplexer. Specify the connections for the select lines, data inputs, and the output.